A counterexample to a lit-only sigma-game conjecture

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Abstract

A move in the sigma-game toggles the neighbors of a vertex; the lit-only game permits such a move only at a lit vertex. The gap for a graph is the largest difference, over starting configurations, between the minimum numbers of lights attainable in these two games. Goldwasser, Wang and Wu conjectured that a connected graph on n vertices with gap greater than $n/2$ must be complete m-partite for $m \equiv 3 \pmod 4$. We refute their Conjecture 5, published in 2011. The complement of the disjoint union of a six-cycle and an isolated vertex has a starting configuration reducible to zero lights by regular moves but to no fewer than four by lit-only moves. A finite closed set of configurations certifies the latter assertion. The counterexample is proved in Lean from the definitions.

The proofs, the Lean formalization, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 The conjecture and the counterexample

We define the gap between the two games, state the conjecture about large gaps, and give a counterexample. A configuration on a finite undirected graph G assigns to each vertex either ON (lit) or OFF (dark). Its weight is the number of lit vertices. A regular move at a vertex toggles all its neighbors. A lit-only move makes the same change only if the played vertex is ON. Loops are allowed in the source convention: the played vertex is its own neighbor exactly when it has a loop. Moves at dark vertices do nothing in the lit-only game and may be omitted.

For a starting configuration x, let $\mathrm{ML}_G(x)$ be the least number of lit vertices attainable by regular moves, and let $\mathrm{ML}^*_G(x)$ be the corresponding lit-only minimum. These minima include the empty sequence of moves. Following Goldwasser, Wang and Wu [1], define

$$\mathcal{D}(G) = \max_x (\mathrm{ML}^*_G(x) - \mathrm{ML}_G(x)).$$

Both minima in each difference use the same starting configuration.

Theorem 1. *Let G be the complement of the disjoint union of the six-cycle $0,1,2,3,4,5,0$ and the isolated vertex 6. Let x have lit vertices $\{0,1,3,4\}$. Then*

$$\mathrm{ML}_G(x) = 0, \qquad \mathrm{ML}^*_G(x) = 4, \qquad \mathcal{D}(G) \geq 4 > \frac{7}{2}.$$

The graph G is connected and is not complete multipartite.

A complete m-partite graph is loopless, with its vertex set partitioned into m nonempty parts, two distinct vertices being adjacent exactly when they lie in different parts. Goldwasser, Wang and Wu's Conjecture 5 [1, p. 5] starts, "Let G be a connected graph on n vertices." Its full implication, in mathematical form, is

$$\mathcal{D}(G) > \frac{n}{2} \quad \Longrightarrow \quad \begin{array}{c} G \text{ is a complete } m\text{-partite graph for some positive integer } m, \\ m \equiv 3 \pmod 4. \end{array}$$

Theorem 1 contradicts that implication even for a loopless graph. The conjecture appears in the article published October 31, 2011 [1].

The contribution is the explicit graph and configuration in Theorem 1, together with a short certificate for the difference between their minimum light numbers. Goldwasser, Wang and Wu studied the parameter and posed the question, supplied the complete multipartite examples motivating it, and reported earlier theoretical and experimental investigations [1, Example 4 and pp. 5–6]. Wang and Wu's 2009 preprint [2] studies related differences, including a difference between two graph-wide maxima; that quantity must be distinguished from $\mathcal{D}(G)$. Wu and Xiang restated the present implication in their 2019 survey and described prior experiments on lit-only reachability [4, Conjecture 3.3 and pp. 10–11 of the author manuscript]. Their investigations form the background to the present question.

Nearby algebraic work includes Wu's analysis of lit-only games on line graphs [3], and Sjöstrand's reachability characterization, first posted in 2023 and published in 2025 [5]. The argument below does not require either theory: its essential lower-bound step is to exclude every lit-only path to a configuration of weight less than four. Exhibiting some paths alone would not do this. A small set closed under every allowed move supplies the exclusion.

Proof overview

Section 2 specifies the adjacency lists and the starting configuration. Two regular moves turn every light off. For the lit-only game, a closed set contains the starting configuration and has no state with fewer than four lights. Induction on the length of a move sequence therefore proves the lower bound, and the starting configuration attains it. The nonedges 01 and 12, together with the edge 02, show that nonadjacency fails transitivity and rule out every complete multipartite graph. Section 3 describes the Lean proof, the certificate's independent checkers, and their scope.

2 The finite certificate

We give all graph and configuration data needed to prove Theorem 1. Figure 1 shows the graph with the starting lights. Its universal vertex 6 makes the graph connected.

For compact exact data, encode a configuration with lit set $A \subseteq \{0, 1, \ldots, 6\}$ by the integer $\sum_{i \in A} 2^i$. Thus the starting configuration has code $27 = 1 + 2 + 8 + 16$. Write r_v for the code of the neighbor set of vertex v. The adjacency data are

$$(r_0, r_1, r_2, r_3, r_4, r_5, r_6) = (92, 120, 113, 99, 71, 78, 63). \tag{1}$$

For example, $r_0 = 92 = 4 + 8 + 16 + 64$ means that vertex 0 has neighbors $\{2, 3, 4, 6\}$. The rows are symmetric and have no diagonal entries. Among vertices $0, \ldots, 5$, exactly the consecutive cycle pairs are missing; vertex 6 is universal. These are precisely the graph in Theorem 1.

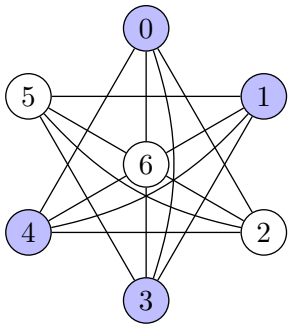


Figure 1: The graph in Theorem 1. Filled vertices are ON in the starting configuration x; open vertices are OFF.

Write $a \oplus b$ for bitwise exclusive or of two nonnegative integer codes. A regular move at v sends z to $z \oplus r_v$. In particular, the regular moves $0, 4$ give

$$27 \xrightarrow{0} 27 \oplus 92 = 71 \xrightarrow{4} 71 \oplus 71 = 0.$$

No configuration has a negative number of lights, so this proves $\mathrm{ML}_G(x) = 0$. The second move in this sequence is unavailable in the lit-only game, because vertex 4 is dark in state 71.

For the lit-only lower bound consider the set of configuration codes

$$S = \{27, 45, 54, 63, 71, 78, 92, 99, 113, 120\}. \tag{2}$$

The starting state belongs to S. Table 1 lists, for each state, the vertices at which a lit-only move is permitted and all their successor states. The successors in every row belong to S.

z	Lit vertices	Successors
27	$\{0, 1, 3, 4\}$	$71, 99, 120, 92$
45	$\{0, 2, 3, 5\}$	$113, 92, 78, 99$
54	$\{1, 2, 4, 5\}$	$78, 71, 113, 120$
63	$\{0, 1, 2, 3, 4, 5\}$	$99, 71, 78, 92, 120, 113$
71	$\{0, 1, 2, 6\}$	$27, 63, 54, 120$
78	$\{1, 2, 3, 6\}$	$54, 63, 45, 113$
92	$\{2, 3, 4, 6\}$	$45, 63, 27, 99$
99	$\{0, 1, 5, 6\}$	$63, 27, 45, 92$
113	$\{0, 4, 5, 6\}$	$45, 54, 63, 78$
120	$\{3, 4, 5, 6\}$	$27, 63, 54, 71$

Table 1: Closure of S under lit-only moves. Successors are in the order of the lit vertices listed in the middle column. Each is obtained by exclusive or with the corresponding row of (1).

Every lit-only state reachable from 27 lies in S. Indeed, this is true before any move; if it holds after a sequence of moves, the next allowed move stays in S by Table 1. This proves the claim by induction. The state 63 has six lights and each of the other states in S has four. Consequently $\mathrm{ML}^*_G(x) \geq 4$. The starting state itself has four lights, so equality holds. Together with the regular minimum this gives $\mathcal{D}(G) \geq 4 > 7/2$.

Finally, 01 and 12 are nonedges of G, whereas 02 is an edge. In a complete multipartite graph, nonadjacent distinct vertices must lie in the same part. The two nonedges would put $0, 1, 2$ in one part, in contradiction to the edge 02. Thus G is not complete multipartite. This completes the proof of Theorem 1.

3 Formalization, computational checks and scope

The Lean project in `lean/` proves Theorem 1 from the game definitions, including the induction over arbitrary finite move sequences. The theorem `LitOnlySigma.counterexample` proves the graph's symmetry and connectivity, the two same-start minima, failure of every complete multipartition, and the negation of Conjecture 5. The direct refutation is also named `LitOnlySigma.not_conjecture5`. The definitions and their correspondence with the source are given in `lean/Statement.lean` and `lean/FAITHFULNESS.md`.

With elan installed, the proof and its axiom guards are checked by running the following commands from the accompanying folder:

```
cd lean
lake exe cache get
lake build
lake env lean Audit.lean
cd ..
```

The toolchain and dependency manifest pin Lean and Mathlib to version 4.34.0. Finite certificate checks use kernel reduction; the proof imports no search program or external checker. The build checks the expanded endpoint types, the graph and starting-configuration descriptions, and a control showing that deleting state 45 destroys closure. Its guarded axiom reports require exactly `propext`, `Classical.choice` and `Quot.sound`; an additional axiom or proof hole fails the build. The audit command replays the axiom check separately. The trusted base is Lean's kernel and these standard logical foundations, together with the stated identification of the definitions with the source conjecture.

The accompanying checkers verify the finite certificate used in the proof. The file `data/certificate.json` contains the adjacency rows (1), the starting code, the two regular moves, and the closed set (2). The public check is

```
python3 checks/check.py
```

It runs two independent exact checkers and corruption controls, and confirms that the graph illustration and closure table agree with the certificate. One checker uses integer bitmasks; the other uses explicit sets of vertices and symmetric difference, and additionally enumerates the configurations reachable from the starting state in each game. Neither checker imports search code. Both reject the corrupted certificate obtained by deleting state 45 from (2). The entry point needs Python 3.10 or later and only its standard library. The small finite data can also be checked directly from (1) and Table 1.

The verified finite obligations are the graph's stated adjacency, connectivity and failure to be complete multipartite; the regular path to zero; and containment of every allowed successor of every listed state in the listed set, with its asserted minimum number of lights. The mathematical passage from closure to all finite lit-only move sequences is the induction given above, also proved in Lean. The checkers use exact integer or set operations; there is no numerical approximation.

The result concerns the exhibited starting configuration and its consequence for the strict-threshold implication. It asserts neither minimality of the example nor an exact graph-wide value of $\mathcal{D}(G)$, and it does not settle the other light-number or maximum-degree conjectures in the cited papers.

References

[1] J. Goldwasser, X. Wang and Y. Wu, *Minimum light numbers in the σ-game and lit-only σ-game on unicyclic and grid graphs*, Electronic Journal of Combinatorics **18**(1) (2011), P214, 27 pp. Published October 31, 2011. doi:10.37236/701.

[2] X. Wang and Y. Wu, *Difference between minimum light numbers of sigma-game and lit-only sigma-game*, arXiv:0904.3050, first posted April 20, 2009. https://arxiv.org/abs/0904.3050.

[3] Y. Wu, *Lit-only sigma game on a line graph*, European Journal of Combinatorics **30**(1) (2009), 84–95. doi:10.1016/j.ejc.2008.02.003.

[4] Y. Wu and Z. Xiang, *The lit-only σ-game: some mathematics behind a puzzle for kids*, Mathematics Student **88** (2019), 67–98. Author manuscript.

[5] J. Sjöstrand, *Orbits under dual symplectic transvections*, Linear Algebra and its Applications **710** (2025), 507–530. doi:10.1016/j.laa.2025.02.010. First posted December 6, 2023, as arXiv:2312.03933: https://arxiv.org/abs/2312.03933.